

# Wenbo Lyu

B.Sc. Graduate

## Curriculum Vitae

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## Education and qualifications

B.Sc. In Geographic Information Science Shaanxi Normal University 2025

## Research Experience

2026.6- Research Assistant, The Hong Kong Polytechnic University  
2025.9-2026.6 Research Assistant, Eastern Institute of Technology, Ningbo  
2024.8-2025.8 Research Intern, The Hong Kong University of Science and Technology (Guangzhou)

## Research interests

- My research interests lie in **advancing methodologies in spatial causal inference** and **developing high-performance computational tools**, with a primary focus on *R packages*.
- Currently, my work centers on **Empirical Dynamic Modeling (EDM)** framework for modeling *dynamic system*, **information theory** for quantifying *information flow* and *variable interactions*, **ordinary differential equations (ODEs)** for characterizing *spatiotemporal processes* and *system evolution*, as well as **counterfactual** and **potential outcomes** framework for estimating *causal effects*. I am particularly interested in leveraging these approaches to address critical challenges in *urban sustainability*, *climate change mitigation*, and broader global issues.
- I specialize in *data analysis*, *statistical modeling*, and developing open-source analytical tools, including *R packages*, using **R**, **C++**, and **Python**, with a strong focus on *spatial analysis*. I actively contribute to the R geospatial community and am dedicated to advancing open-source geospatial software.

## Selected research papers

1. Lyu, W, Dai, S, Song, Y, Zhao, W, Yi, W, Xiao, Y, and Jia, N (2026). Measuring causal strengths from spatial cross-sectional data with geographical cross mapping cardinality. *International Journal of Geographical Information Science*, 1–23. DOI: 10.1080/13658816.2026.2687121.
2. Lyu, W, Lei, Y, Yi, W, Song, Y, Li, X, Dai, S, Qin, Y, and Zhao, W (2026). Causal discovery in urban data with temporal empirical dynamic modeling: The R package tEDM. *Computers, Environment and Urban Systems* 127, 102435. DOI: 10.1016/j.compenvurbsys.2026.102435.
3. Lv, W, Lei, Y, Liu, F, Yan, J, Song, Y, and Zhao, W (2025). gdverse: An R Package for Spatial Stratified Heterogeneity Family. *Transactions in GIS* 29(2), e70032. DOI: 10.1111/tgis.70032.
4. Lv, W, Liu, F, Cai, K, Cao, Y, Deng, M, Liang, W, Yan, J, and Wang, G (2024). Distinguishing the Impacts and Gradient Effects of Climate Change and Human Activities on Vegetation Cover in the Weihe River Basin, China. *Journal of Geophysical Research: Biogeosciences* 129(10), e2024JG008297. DOI: 10.1029/2024jg008297.
5. Li, B, Lv, W, Ding, H, Song, Y, and Zhao, W (2026). New-type urbanization policy, urban expansion, and employment quality in chinese cities: evidence from a quasi-natural experiment in China. *GIScience & Remote Sensing* 63(1), 2716413. DOI: 10.1080/15481603.2026.2716413.
6. Xu, L, Shi, H, Lyu, W, Zhang, Y, and Zhao, H (2026). Beyond correlation: Unveiling the causal link between street-scapes and recreational walking/cycling. *Journal of Transport Geography* 136, 104789. DOI: 10.1016/j.jtrangeo.2026.104789.
7. Chen, C, Song, Y, Lv, W, Shemery, A, Hampson, K, Yi, W, Zhong, Y, and Wu, P (2025). Predicting pavement cracking performance using laser scanning and geocomplexity-enhanced machine learning. *Computer-Aided Civil and Infrastructure Engineering*. DOI: 10.1111/mice.13489.

Note: Publications under “Lyu, W.” and “Lv, W.” both refer to Wenbo Lyu.

## Unpublished working papers

Quantifying pairwise directional influence between spatial variables: A reconstruction-based approach to asymmetric relationships  
Spatial Pattern Entropy: A Symbolic Measure for Continuous Spatial Data  
Spatially convergent partial cross mapping for identifying direct causal influence in cross-sectional data  
Unveiling interaction between spatial processes with information decomposition  
Pattern Causality for Detecting Interactions with the R Package pc  
coupling: Coupling Coordination Analysis in R  
Measuring spatial stratified heterogeneity via information imbalance  
Spatial causal discovery under latent confounders via deep orthogonal representation learning

## Selected awards and honours

- 2024 Longi Non-Education Major Scholarship
- 2024 First Prize in the 13th National University Student GIS Application Skills Competition
- 2023 Grand Prize in the 12th National University Student GIS Application Skills Competition
- 2023 First Prize in the Second National University Student Ecological Environment Management Research Innovation Competition
- 2023 Second Prize of the 5th 'Guodi Cup' National College Student Natural Resource Science and Technology Competition
- 2021 Outstanding Individual in Military Training Publicity for College Students, Shaanxi Normal University

## Software (R packages)

- spEDM Spatial Empirical Dynamic Modeling
- tEDM Temporal Empirical Dynamic Modeling
- pc Pattern Causality Analysis
- gdverse Analysis of Spatial Stratified Heterogeneity
- sdsfun Spatial Data Science Complementary Features
- coupling Coupling Coordination Analysis
- infoxtr Information-Theoretic Measures for Revealing Variable Interactions